MONASH UNIVERSITY · ENGINEERING PROJECT

Episodic Memory for Robots in the Wild

Benchmarking VLM recall in long, event-sparse egocentric dashcam drives

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SPARSE + TEMPORAL PIPELINES COMPLETE · MODEL EVALUATION IN PROGRESS

The client's hard case is long, quiet and repetitive

TYPICAL VIDEO QA

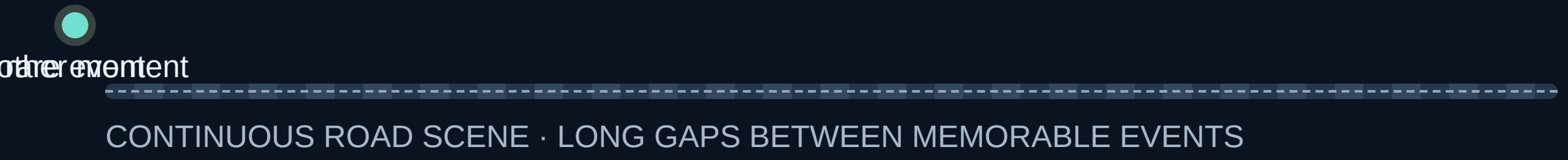
Many visible cues

Short clips, frequent actions and distinctive scenes make it easier to answer from local visual evidence.

OUR TARGET SETTING

Few events across a long drive

Forward-facing vehicle video in forests, rural roads and tunnels; target durations are roughly 6 minutes to 1 hour.



Client need: test whether an agent can verify what occurred and when—without guessing from familiar driving patterns or inventing plausible events.

Turn episodic memory into four measurable capabilities

Research question: can a VLM distinguish real from absent events, remember their order, and reason over observations spread across a long drive?

01 · SPARSE EVENT LOCALISATION

Did the rare event happen?

Existence, noteworthy-moment and deceptive-question tests.

PIPELINE IMPLEMENTED

02 · TEMPORAL ORDER

What happened first, next or before?

Tests chronology across events separated in time.

PIPELINE IMPLEMENTED

03 · SPATIAL REASONING

Where, and how far?

Grounded location and distance reasoning with few landmarks.

PLANNED

04 · COUNTING

How many distinct objects appeared?

Tracks objects across the drive, not just within one frame.

PLANNED

Three delivered extensions make the benchmark usable

EXTENSION 01

Sparse-event benchmark

Engineering: section-level detection, duplicate review and questions with plausible false events.

Client value: measures both missed events and unsupported claims.

EXTENSION 02

Temporal-order benchmark

Engineering: timestamp-grounded event timelines, human review and order-sensitive questions.

Client value: separates remembering the event from remembering its place in the drive.

EXTENSION 03

Cross-model QA + grading

Engineering: configurable model inputs, answer parsing and automatic exact-match scoring.

Client value: repeatable comparisons and traceable per-question failures.

Together, the extensions move the work from a concept toward an auditable evaluation workflow.

Ground truth is built with timestamps and a human review gate

01 · SEGMENT

Timestamped frames

Sample into model-sized sections; keep true elapsed-time ranges in a manifest.

→

02 · EXTRACT

VLM observations

Task prompts ask for rare events or a section-level event timeline.

→

03 · RECONCILE

Order + merge

Python sorts by frame timestamp; the LLM merges duplicates, without inventing or reordering.

→

04 · VERIFY

Human review

Check each event against its frame evidence before generating questions.

Sparse output: reviewed `events.json` → existence, deceptive and noteworthy questions.

Temporal output: reviewed `timeline.json` → pairwise, sequence and before/after questions.

Question types probe recall, not just recognition

SPARSE EVENT LOCALISATION

Can the model resist a plausible guess?

- **Existence:** did a verified event occur?
- **Deceptive:** did a plausible but absent event occur? Correct answer: no.
- **Noteworthy:** choose the one verified event among four candidates.

TEMPORAL ORDER

Can the model recover chronology?

- **Pairwise:** which of two events happened first?
- **Sequence:** put three to five observed events in order.
- **Before/after:** identify what immediately preceded or followed an event.

Each generated item retains event IDs and frame evidence for review; the answering model receives the question and options, not the answer key or evidence metadata.

One questionnaire, configurable model inputs, automatic grading

1 · BLIND THE QUESTIONS

Hide the key

Remove answers and evidence from the prompt; randomize question order.

2 · RUN THE MODEL

Record the input

Use bounded, uniformly sampled frames or a supported native-video path; save input metadata.

3 · SCORE THE OUTPUT

Compare answer indices

Exact-match grading; sequence questions are order-sensitive.

answers.json
model response + input metadata

graded_results.json
per-question correctness + accuracy

Compare by condition
model × task × drive

The benchmark pipelines are ready; model scores are still running

BENCHMARK CREATION

Sparse + temporal stages

Extraction, review/timeline construction and question generation are implemented.

COMPLETE

EVALUATION TOOLING

Questionnaire + grader

Model input adapters, answer parsing and exact-match grading are implemented.

READY

CURRENT EXPERIMENT

Answering across models

The model question-answer runs are underway; final comparisons are pending.

IN PROGRESS

No measured model leaderboard is shown yet. The next evidence will be per-task scores plus reviewed examples of correct answers, misses and hallucinations.

Spatial reasoning and counting remain future implementation work.

Make the next results decision-ready

LIMITATIONS & TRADE-OFFS

Interpret scores carefully

- Model-assisted event extraction still needs human verification.
- Frame sampling and context limits can miss brief events.
- Different native-video and sampled-frame inputs can affect comparability.
- Spatial and counting tasks are not yet implemented.

RECOMMENDED NEXT STEPS

Turn runs into evidence

- Finish model runs; report accuracy by task, drive and input condition.
- Audit false positives, missed events and order errors with frame evidence.
- Expand scene coverage and implement spatial reasoning/counting.

Recommended reporting: matched input budgets, per-question error examples and a clear separation between measured results and projections.

A repeatable stress test for long-horizon memory

What the client gets today: traceable sparse-event and temporal-order evaluations built for long, low-signal driving video—plus a reusable path to compare VLMs.

NEXT MILESTONE

Complete the comparison

Finish model runs → publish task-level plots and failure examples → prioritize the next benchmark extension.

CLIENT DISCUSSION

Which failure is most costly?

Missing a rare event, inventing one, or remembering it in the wrong order?